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RESEARCH

Acute:chronic workload ratio of professional soccer players preceding hamstring muscle injuries: a 2-season retrospective study

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Abstract

Purpose Given the controversial use of the Acute:Chronic Workload Ratio (ACWR) in determining the availability of soccer players for training and matches, this study aimed at verifying how many professional soccer players had an ACWR above 1.5 (considered a 'danger zone') immediately before the exposure (training or match) in which they sustained a hamstring muscle injury.

Methods Forty-eight male soccer players of a professional team were monitored through global positioning system (GPS) in this 2-season retrospective study (10 participated in both seasons and 38 in a single season). The ACWR of total distance, high-speed running distance, very high-speed running distance, and the number of actions performed > 19 km/h were calculated before all hamstring injuries and included in the analysis.

Results Twenty players sustained hamstring injuries. There was no time effect on the group's daily workload or the average workload over the 4 weeks before the injury ($p > 0.05$). Fifteen players (i.e., 75%) had ACWR below 1.5 for the four GPS metrics immediately before the training or match in which they sustained a hamstring injury. Five players (25%) were classified in the 'danger zone' (i.e., ACWR > 1.5) for the very high-speed running distance; 3 of these players were also in the 'danger zone' for high-speed running distance and the number of actions > 19 km/h, and 2 players for the total distance.

Conclusion Since three-quarters of the hamstring injuries affecting professional soccer players were preceded by ACWR values below 1.5 in all GPS metrics, it is prudent for medical/coaching staff exercise caution when considering the ACWR 'danger zone' as the primary metric for determining player availability for training or matches.

Keywords Knee flexors · Muscle injury · Injury prevention · Football

Abbreviations

ACWR	Acute:chronic workload ratio
GPS	Global positioning system
OR	Odds ratio
RR	Relative risk
UEFA	Union of European Football Association

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Introduction

The physical demands on professional soccer players have increased over the years. A top-level European team played around 20% more matches in the 2018/2019 season compared to the 2008/2009 season [1]. Data from the English Premier League reveal a modest increase of ~ 2% in the total distance covered during a match (~ 11 km) from season 2006/2007 to season 2012/2013, while the running distances covered in high-speed and very high-speed increased by ~ 29% and ~ 51%, respectively [2]. Hence, matches have been played with escalating frequency and intensity, a trend expected to persist in the coming years [1]. The potential stress due to the increased physical demands raised concerns within the sports medicine community. Injuries can have adverse effects on players' careers [3], team performance [4], and club finances [5]. Consequently, elite soccer clubs continually invest in strategies aimed at mitigating the burden of injuries [6, 7].

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Monitoring athletes' workload has been a longstanding practice in the development of periodization plans aimed at enhancing physical performance [8]. More recently, workload monitoring has also been recommended as a pivotal strategy for preventing sports injuries [9]. This presents a challenge in team sports like soccer, where the simultaneous monitoring of a typical 30-player squad is required. Measures of workload have been categorized as either internal or external. Internal workloads relate to relative biological stressor (both physiological and psychological) imposed on the athlete, such as heart rate, blood lactate and ratings of perceived exertion. External workloads are objective measures of the work performed by the athlete [10], including parameters provided by the global positioning system (GPS) and accelerometry. The total distance, the distance covered in specific speed ranges, and the number of specific actions (e.g., sprints, accelerations) are commonly analyzed in studies focusing on the association between external workload and injury risk [11].

In the last decade, studies with different sports, including professional soccer [12, 13], have reported a positive association between sudden increases in the athletes workload, commonly referred to as 'spikes', and risk of injuries. This has led to a resurgence of interest in the 'acute:chronic workload ratio' (ACWR) as a valuable tool for monitoring these workload spikes [14]. This measure describes the ratio between the 'acute workload' (workload of the most recent week) and the 'chronic workload' (average workload of the previous 2–8 weeks) [11]. Thus, a higher chronic workload (that is, the athlete has developed 'fitness') builds the foundation to support acute workload (i.e., the 'fatigue'). Otherwise, if the acute workload exceeds the chronic workload, the athlete may fatigue which could lead to an increased risk of injury [15]. ACWR values between 0.8 and 1.3 have been classified as a 'sweet spot', theoretically lowering the risk of injury, while athletes with ACWR values above 1.5 have been classified as in a 'danger zone' due to the increased likelihood of injury [15].

While the ACWR's popularity may stem from its practicality in helping practitioners manage undesirable workload spikes, it is reasonable to hypothesize that a single cut-off point may not be suitable for all workload parameters, injury types, and sports. Furthermore, the ACWR has faced criticism in recent times [16, 17], partly because of conflicting findings regarding the relationship between ACWR GPS metrics and injury risk in soccer studies [12, 18]. Although elevated ACWR values do not necessarily forecast immediate injury, practitioners are confronted with the quandary: '*Should players with high ACWR be prevented from participating in training or matches to reduce the risk of injuries?*' This retrospective study addressed this issue on the most prevalent non-contact injury in professional soccer: the hamstring muscle injury [19]. Our goal was to verify how many professional soccer players had an ACWR above 1.5 (referred to as the 'danger zone') immediately before the exposure (training or match) in which they sustained a hamstring muscle injury. This information aids coaching and medical staff in their decision-making process regarding participation of players in training or matches.

Methods

Study design

This is a two-season retrospective study. A soccer club agreed to furnish data comprising 28 days of the GPS external workload from those players who sustained hamstring muscle injuries during the 2018 and 2019 seasons. The approval to conduct the study was granted by the club board, medical and coaching staff. The data included in the present study originated from a routine physical evaluation performed by the club. The study was approved by the University's ethics committee (#2.390.889). Due to the retrospective nature of the analysis without interfering with the routine of the participants, signatures of the informed consent form were not required [20]; however, a coaching staff member surveyed the recorded data of injured athletes and extracted their GPS data without personal identification to ensure confidentiality.

Participants

This study involved professional male soccer players from a single soccer club. The team competed in the Brazilian Championship Series A and B in the first and second season of the study, respectively. Along the 2-season period, all team squad players were monitored during field training sessions and matches with GPS technology to collect horizontal displacements (distance, time, and velocity). GPS data of all players who sustained a hamstring injury during the study period were considered for analysis. Only players who trained and played regularly with no recent injury history were included in the analyses. Hamstring injuries were confirmed by the club medical doctor through clinical examination and imaging exams. Details about each hamstring injury were also reviewed, including the mechanism of the injury (e.g., running, stretching), the side (dominant or non-dominant limb), the circumstance (training or match), the classification (functional or structural; according to the Munich consensus statement) [21], and the severity (minimal, mild, moderate and severe) [22]. The coaching staff gathered and supplied us with anthropometric data, including body

mass measured in the week of the injury, height, and age, as well as player positions categorized as defenders, midfielders, and forwards.

GPS external workload

Multiple external workload metrics were quantified using a GPS-based tracking system (Polar Team Pro System Pro; Polar Electro, Kempele, Finland). As specified by the manufacturer, the system combines signals from a 10 Hz GPS with a 200 Hz microelectromechanical inertial measurement unit consisting of a triaxial accelerometer, gyroscope, and magnetometer. Data were downloaded into Polar Team Pro analysis software (Polar Electro, Kempele, Finland). The following workload parameters were analyzed: total distance (includes walking, jogging, running and sprinting; in meters), high-speed running distance (distance covered between 19.1 and 22.9 km/h; in meters), very high-speed running distance (distance covered over 23 km/h, in meters), and the number of actions performed at highest speed zones (actions > 19 km/h) in each player exposition (training and matches). On days with multiple expositions (e.g., two training sessions), data were summed to compute the daily workload. On days with no GPS monitored sessions (for example, training sessions in the gym or day offs), the workload was considered zero for all metrics. There have been no reported instances of GPS data loss among the players included in the study.

GPS data from the four weeks preceding each hamstring muscle injury occurrence were analyzed. The 28-day workload data were retrospectively categorized from the date of the hamstring injury event ("day 0"). Daily external workload (from day 1 to day 28 before the injury), weekly workload (sum of 7 consecutive days, from week 1 to week 4 before the injury), acute workload [total workload of the 7 days before the injury (week 1)], and chronic workload (average workload of weeks 2 to 4) were calculated. As recommended by Lolli et al. [23], we did not include the acute workload period in the calculation of the chronic workload. Lastly, the acute workload was divided by the chronic workload to calculate the ACWR.

Statistical analysis

The results are presented in mean values, standard deviations and 95% confidence intervals for continuous variables, and absolute and relative frequencies for categorical variables. Between-days and between-weeks workloads were compared through repeated measures ANOVA, followed by Bonferroni post-hoc test when there was a significant time effect. The Pearson coefficient (r) and the R-squared (R^2) were calculated for the correlations and the coefficient of determination, respectively, between the workload and days before the injury. Statistical significance was set at $p < 0.05$. The statistics were performed using the JASP 16.0 software.

Results

Forty-eight players were followed in this study (10 participated in two seasons and 38 in a single season). Twenty players suffered hamstring muscle injuries. Table 1 describes the characteristics of players and hamstring injuries.

There was no tendency to increase or decrease the players' daily workload over the 4 weeks before the injury: total distance [$F = 0.64$, $p = 0.91$; $r = 0.05$ (−0.34–0.46), $p = 0.76$; $R^2 < 0.01$]; high-speed running distance [$F = 0.80$, $p = 0.74$; $r = 0.13$ (−0.26–0.53), $p = 0.48$; $R^2 = 0.01$]; very high-speed running distance [$F = 0.50$, $p = 0.98$; $r = -0.20$ (−0.59–0.19), $p = 0.29$; $R^2 = 0.04$]; and number of actions > 19 km/h [$F = 0.71$, $p = 0.84$; $r = 0.16$ (−0.23–0.56), $p = 0.39$; $R^2 = 0.03$] (Fig. 1). There was no statistical difference for weekly average workload (Table 2): total distance ($F = 0.17$, $p = 0.91$); high-speed running distance ($F = 0.48$, $p = 0.69$);

Table 1 Characteristics of players and hamstring muscle injuries [mean ± SD or frequency (% of total)]

Age	24.6 ± 3.6
Body mass (kg)	75.9 ± 5.9
Height (m)	1.76 ± 4.1
Players' position	
Forwards	9 (45%)
Midfielders	4 (20%)
Defense	7 (35%)
Injured limb	
Dominant	12 (60%)
Non-dominant	8 (40%)
Injury circumstance	
Training	10 (50%)
Match	10 (50%)
Injury mechanism	
Running	15 (75%)
Stretching	2 (10%)
Jumping	1 (5%)
Unknown	2 (10%)
Injury classification	
Functional	7 (35%)
Structural	13 (65%)
Injury severity	
Minimal	1 (5%)
Mild	4 (20%)
Moderate	14 (70%)
Severe	1 (5%)

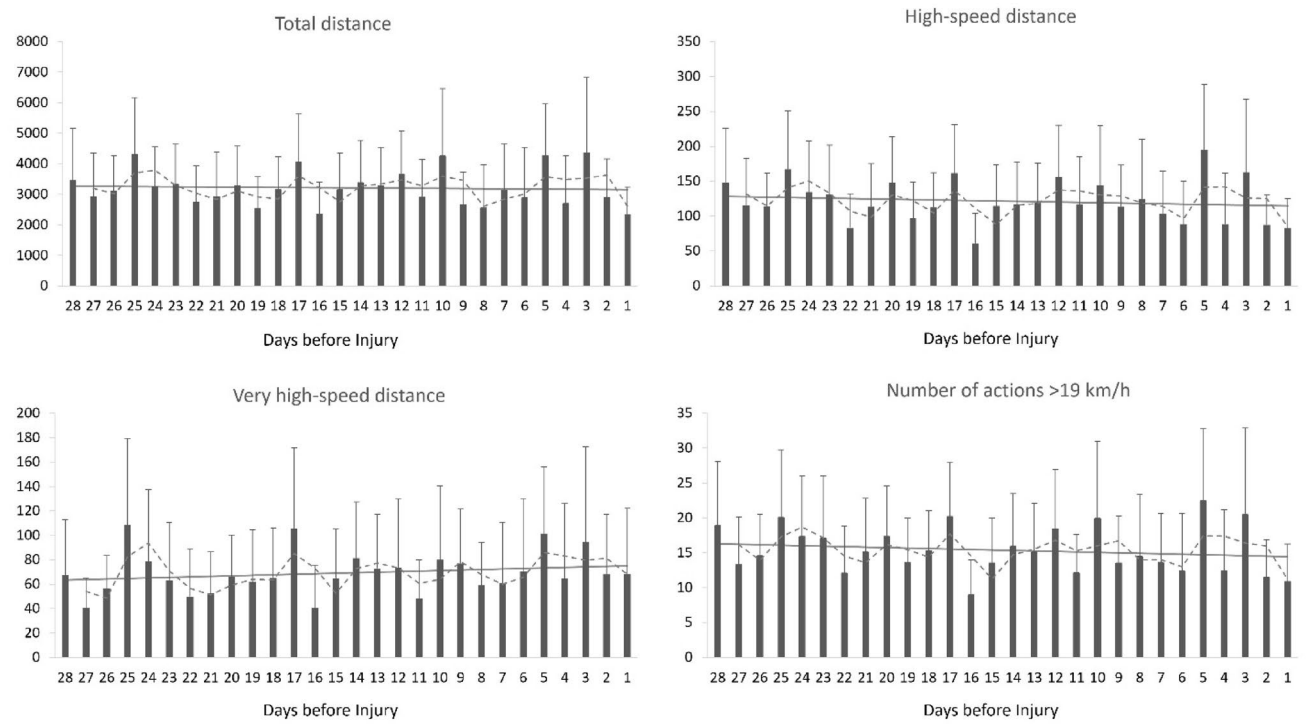

Fig. 1 Charts illustrate the average daily external workloads along the 28 days before de hamstring injury. Day 1 (right on the *x*-axis) is the day before the hamstring muscle injury. Distance is presented in meters. The bars represent the group averages for the given day and error bars represent the upper 95% confidence interval. The gray horizontal solid lines represent the trendlines and the dashed lines represent the moving average along the 28 days

Table 2 Weekly workload and total workload of the 4 weeks before hamstring muscle injuries [mean ± SD (95%CI)]

	Chronic workload			Acute workload	
	Week 4	Week 3	Week 2	Week 1	Total (sum)
Total distance (km)	23.1 ± 9.5 (13.1–33.0)	21.4 ± 7.1 (12.8–30.1)	22.7 ± 4.4 (12.7–32.6)	22.5 ± 11.2 (11.5–33.6)	89.8 ± 22.3 (80.0–99.6)
High-speed running distance (m)	892.2 ± 447.7 (423.9–1,360.4)	807.2 ± 320.2 (405.8–1,208.6)	890.5 ± 256.8 (399.1–1,381.8)	808.2 ± 411.1 (330.1–1,286.2)	3,398.1 ± 995.2 (2,961.9–3,834.3)
Very high-speed running distance (m)	464.8 ± 319.1 (152.3–777.3)	456.4 ± 259.2 (162.7–750.0)	491.8 ± 230.6 (173.0–810.6)	528.2 ± 340.3 (122.7–933.6)	1,941.3 ± 843.1 (1,571.7–2,310.8)
Number of actions > 19 km/h	113.2 ± 54.2 (57.0–169.4)	103.8 ± 40.0 (57.0–150.5)	109.4 ± 30.0 (54.1–164.7)	103.6 ± 48.9 (45.9–161.2)	430.1 ± 123.3 (376.0–484.1)

very high-speed running distance ($F = 0.39$, $p = 0.75$); and number of actions > 19 km/h ($F = 0.32$, $p = 0.80$).

As illustrated in Fig. 2, the group's ACWR mean values were: total distance, 1.08 ± 0.62 (0.80–1.35); high-speed running distance, 1.02 ± 0.69 (0.72–1.33); very high-speed running distance, 1.30 ± 1.27 (0.74–1.86); number of actions > 19 km/h, 1.03 ± 0.63 (0.75–1.31). When excluding a single outlier case for very high-speed running distance (who has ACWR of 5.84 for this variable), the ACWR for very high-speed running distance was 1.07 ± 0.72 (0.74–1.39), with a confidence interval below the 'danger zone' (ACWR above 1.5, red zone in Fig. 2).

Considering the individual ACWR status, five players (25% of all players) were classified in the 'danger zone' (ACWR above 1.5, red zone in Fig. 2) for the very high-speed running distance. Three of these players were also in the 'danger zone' for the high-speed running distance and the number of actions > 19 km/h, and two players for the total distance (Fig. 2).

Players with ACWR above 1.5 were 24.8 ± 4.4 (20.9–28.7) years old, while players with ACWR below 1.5 were 24.5 ± 3.5 (22.8–26.4) years old. Hamstring injuries occurred during matches for 40% and 55% of players with ACWR above or below 1.5, respectively. The absence days

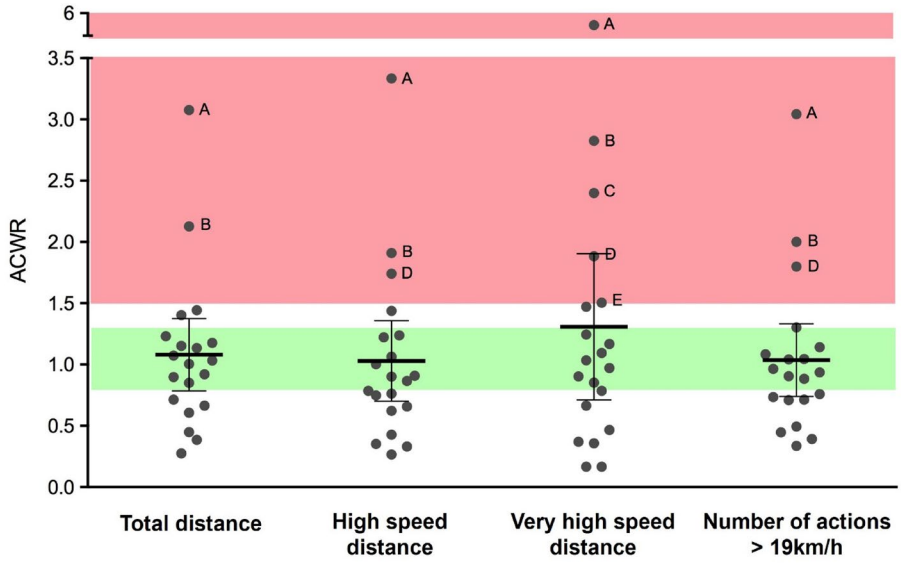

Fig. 2 Acute:chronic workload ratio (ACWR) immediately before the training/match in which players sustained hamstring muscle injuries. Gray circles denote individual values. Thick and narrow lines represent means and 95% confidence intervals, respectively. The 'sweet spot' is dashed in green (ACWR from 0.8 to 1.3), and the 'danger zone' is dashed in red (ACWR above 1.5). Each letter represents a specific player at the 'danger zone' in one or more GPS metrics

after injury were 12.6 ± 2.3 (10.6–14.6) and 13.1 ± 10.0 (7.8–18.4) for players with ACWR above and below 1.5, respectively.

Discussion

This 2-season retrospective study analyzed the ACWR derived from different GPS metrics of 20 professional male soccer players on the day they suffered hamstring muscle injuries. Our primary finding is that 75% of the players stepped onto the soccer field with ACWR values below 1.5 (i.e., outside the 'danger zone') for all GPS metrics on the day of the injury occurence.

The ACWR gained prominence following a narrative review by Gabbett [15] and has been boosted by an International Olympic Committee consensus recommending it in the management of athletes' training routines [24]. The idea of having a singular metric that enables sports scientists, coaches, and clinicians to assess the balance between 'fitness' and 'fatigue' status, potentially reducing the risk of injury, is appealing. However, it has recently faced substantial criticism [16, 17]. Briefly, it has been claimed that ACWR studies lack a sound conceptual framework, that ACWR measure violates mathematical premises regarding the construction of the ratios by introducing noise and statistical artifacts, and that studies intended to be predictive often prove to be merely descriptive [16, 17]. Nonetheless, the routine practice of evaluating how close players are to the ACWR 'danger zone' has become common in numerous soccer clubs. This widespread utilization of the ACWR within the realm of professional soccer encouraged us to conduct this study.

Although two recent reviews including several sports showed a trend that ACWR values of 0.80 to 1.30 appear to be related to a lower risk of injury [11, 25], the association between external workload and soccer players' injury risk has presented conflicting results [12, 18, 26]. Youth players with an ACWR above 1.76 for total distance and with 1.41 to 1.96 for high-speed running distance were at higher risk of contact (RR = 4.98) and non-contact (RR = 2.55) injuries, respectively, when combined with low chronic workloads [26]. For professional soccer players, the ACWR above 1.18 for high-speed running distance represented a higher injury risk (OR = 1.71) according to Jaspers et al. [18], but not for Bowen et al. [12]. In contrast, Bowen et al. [12] found a high overall injury risk for total distance, mainly for non-contact injury (ACWR above 2.14; RR = 3.67), and a very high overall risk for distances covered at low-intensity (ACWR above 2.15; RR = 2.56). Conversely, Enright et al. [27] observed that nearly all of the 142 non-contact injuries analyzed in European leagues teams occurred with players within the "sweet spot" zone. While evidence directly linking GPS metrics-based ACWR to hamstring injuries in professional soccer is lacking, large and sudden increases in high-speed running distance have been cautioned against due to its association with the increased risk of hamstring injuries in Australian football players [28]. Interestingly, the ACWR related to very high-speed running was the metric with the largest number of players within the 'danger zone' in the present study, suggesting that monitoring intense running efforts might add some value to screening players with increased risk of hamstring injury.

Given that three-quarters of the players were outside the 'danger zone' for the four GPS metrics, our study appears to contradict the popular associative model between ACWR and the risk of hamstring injuries. It could be argued that

25% of the injuries observed in the present study might have been prevented had the club employed the ACWR 'danger zone' as a criterion to determine whether a player was fit for participation. However, considering the complex interaction web of modifiable and non-modifiable factors associated with hamstring injuries [29], a given player removed from a match due to an ACWR above 1.5 could potentially experience a hamstring injury as soon as he returned to field activities following normalization of his ACWR value. Therefore, asserting that 25% of the injuries in the study would have been avoided if the ACWR rule had been adhered to appears overly simplistic and premature. In addition, several other players presumably have been exposed in training and/or matches with ACWR above 1.5 along the 2-season period without sustaining any hamstring injury. Consequently, it is reasonable to assume that the adoption of the ACWR-based criterion would reduce the availability of players for the coach. Combining the knowledge that there is a relationship between player availability and team performance [4] with the lack of a solid scientific basis on the ACWR preditive value for injuries, it seems arbitrary and ultimately imprudent in the context of professional soccer to use this metric as the only criterion for selecting a player for a match.

Interestingly, as illustrated in Fig. 2, we found eight players, which constitutes 40% of the 20 injured players, with ACWR below the 'sweet spot' (i.e., below 0.8) in at least one GPS metric. This implies that more players sustained hamstring injuries under a low acute workload condition compared to their counterparts with a high acute workload. These findings corroborate a previous study with Australian football players in which various low-ACWR risk conditions were identified [30]. Presumably, a sudden discontinuation of chronic workloads in a sport with high physical demands would be conducive to an increased risk of injury [16]. Among various hypotheses, the situation where the ACWR denominator is lower than the numerator could suggest not only a gradual physical deconditioning process rendering athletes susceptible to the high demands of the match but also the possibility that a week of deliberate workload substantial reduction following weeks of excessive workload might not be adequate for recovery. In this context, removing players with ACWR below 0.8 from matches also seems reasonable. Nevertheless, when restricting the participation to only those players within the 'sweet spot,' how many will remain available for the coach? From a practical standpoint, it can be challenging to justify to a coach or manager that a player will not play a match because he is 'too rested' this week (i.e., experiencing a low acute workload).

This study has some limitations. First, as commonly found in field studies and applied in professional soccer settings, the calculation of ACWR did not account for the workload generated by gym-based training. Although athletes typically do not engage in substantial running volume or intensity in the gym, the demand generated by strength exercises can impact a player's fitness and fatigue levels. Second, the club agreed to provide retrospective GPS data exclusively for the injured players, limited to a maximum of 28 days preceding the injury event. Consequently, we were unable to calculate ACWR using longer periods of chronic workload or to make comparisons with the same metrics for players who remained injury-free during the same timeframe. Third, we lacked access to players' internal workload data (e.g., rating of perceived exertion), which could have assisted in comprehending the physiological impact of external workloads at specific times before the hamstring injuries. These limitations did not undermine the core research question regarding the number of players who stepped on the field with ACWR values above 1.5 when the hamstring injuries occurred.

Conclusion

Three-quarters of the hamstring muscle injuries examined in this study occurred when professional soccer players stepped on the field with ACWR values below 1.5. These findings suggest that medical/coaching staff should exercise caution when considering the ACWR 'danger zone' as the primary metric for determining player availability for training or matches.

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Author contributions TC and AH collected the data. All authors analysed the data. JBRA and BMB wrote the main manuscript text and prepared figures 1 and 2. All authors reviewed the manuscript.

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Declarations

Conflict of interest The authors have no conflicts of interest to declare.

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